

A SIMPLE AVERAGE NODAL PRESSURE TETRAHEDRAL ELEMENT FOR INCOMPRESSIBLE AND NEARLY INCOMPRESSIBLE DYNAMIC EXPLICIT APPLICATIONS

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SUMMARY

This paper presents a simple linear tetrahedron element that can be used in explicit dynamics applications involving nearly incompressible materials or incompressible materials modelled using a penalty formulation. The element prevents volumetric locking by defining nodal volumes and evaluating average nodal pressures in terms of these volumes. Two well-known examples relating to the impact of elasto–plastic bars are used to demonstrate the ability of the element to model large isochoric strains without locking.
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KEY WORDS finite elements; explicit dynamics; near incompressible deformation; large strains; tetrahedron

1. INTRODUCTION

Dynamic explicit finite element codes are commonly used for the simulation of large strain impact problems by the automotive, aerospace and military industries.^{1–3} In these codes the 8-noded hexahedron element is typically used to model solid components. The element is under-integrated with a single Gauss point to avoid volumetric locking as most large strain applications of practical interest, such as plastic flow, take place under incompressible or nearly incompressible conditions. Generating three-dimensional meshes using hexahedral elements is a demanding task which at present can only be achieved for simple or moderately complex geometries that can be decomposed into simpler ones. Although this may be sufficient for most simple engineering components, occasionally there may be regions that cannot be meshed without excessive simplifications or that lead to very distorted hexahedral elements. In addition, some applications such as human body impact modelling involve geometries that are far too complex to be meshed using hexahedra.^{4,5} Even when the initial geometry is simple, the presence of very large strains may lead to poorly shaped elements unless some form of adaptive remeshing is used. At present the possibilities of using such adaptive techniques with hexahedral elements are very limited.

A number of powerful three-dimensional unstructured tetrahedral mesh generators and related mesh adaptivity techniques have recently been developed for computational fluid dynamics

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applications.⁶ These codes are either based on the advancing front or on the Delauney techniques and are capable of producing a tetrahedral mesh for any geometry, however complicated. Unfortunately, the standard formulation for tetrahedral elements is known to exhibit volumetric locking for nearly incompressible deformations. In addition some of the correction methods available in the literature based on linear pressure interpolation and bubble functions or similar techniques are unsuited for explicit dynamic applications either because they are too expensive or because they do not lead to an explicit pressure evaluation.⁷⁻⁹

This paper presents a simple technique that allows the use of linear tetrahedral elements even in large strain incompressible situations. The technique is based on the definition of nodal volumes leading to a nodal pressure which is then averaged over the elements. The resulting formulation can be very easily implemented in any explicit code with minor changes to the standard tetrahedral element equations. It must be emphasized, however, that the use of such elements is computationally costly as, typically, the number of tetrahedra is more than five times larger than the number of hexahedral elements needed for the same number of degrees of freedom. Consequently, the proposed elements should only be used for applications where employing a standard hexahedral mesh is impractical.

2. AVERAGED NODAL PRESSURE TETRAHEDRON

Consider a standard 4-noded tetrahedron element as shown in Figure 1. The initial geometry of an element (e) is interpolated in the usual way in terms of the shape functions N_a , $a = 1, \dots, 4$, and the isoparametric co-ordinates ξ_1 , ξ_2 and ξ_3 as

$$\mathbf{X}^{(e)} = \sum_{a=1}^4 N_a(\xi_1, \xi_2, \xi_3) \mathbf{X}_a \quad (1)$$

2.1. Mass lumping and nodal volumes

The new element procedure introduced below will be based on the definition of total volumes associated with each node. To motivate this definition, it is useful to consider first the lumped mass matrix typically employed in explicit dynamic applications. In such applications, the

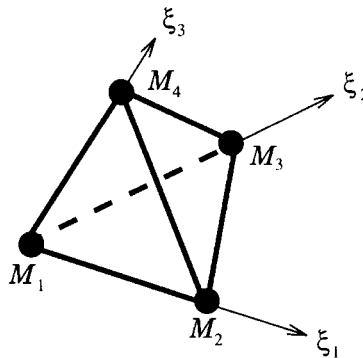


Figure 1. Standard isoparametric tetrahedron

governing equilibrium equations at any given time are enforced via a nodal equilibrium equation, which for a given mesh node a is generally expressed as

$$M_a \mathbf{a}_a = \mathbf{F}_a - \mathbf{T}_a \quad (2)$$

where $\mathbf{F}_a$ represents the equivalent nodal external force; $\mathbf{T}_a$ denotes the equivalent nodal internal force due to the state of stress in the body; $\mathbf{a}_a$ is the acceleration vector of node a and M_a represents the assembled lumped mass at this node. This mass term is typically evaluated by adding the contributions from elements $e = 1, \dots, m_a$ surrounding node a . Each elemental contribution is obtained by integrating the shape function corresponding to node a over the mass of the element to give the total mass associated with node a as

$$M_a = \sum_{e=1}^{m_a} M_a^{(e)}, \quad M_a^{(e)} = \int_{V^{(e)}} \rho_0 N_a \, dV = \frac{1}{4} \rho_0^{(e)} V^{(e)} \quad (3a,b)$$

Removing the density from equations (3) enables the total volume attached to node a to be defined as

$$V_a = \sum_{e=1}^{m_a} V_a^{(e)}, \quad V_a^{(e)} = \frac{1}{4} V^{(e)} \quad (4a,b)$$

Figure 2(a) illustrates this equation for a node of a simple 2-dimensional mesh of linear triangles. The area of each triangle surrounding the central node has been divided into three equal sections by using the element centroids and mid-side points. The sections corresponding to the central node are then joined together to give the shaded polygon shown in the Figure. Clearly, this process is dependent upon the mesh topology, as seen in Figure 2(b), where different nodal connectivities lead to the modified nodal volume shown. It is worth noting that in 3D this dependency, combined with the fact that it is impossible to mesh any region of space with a

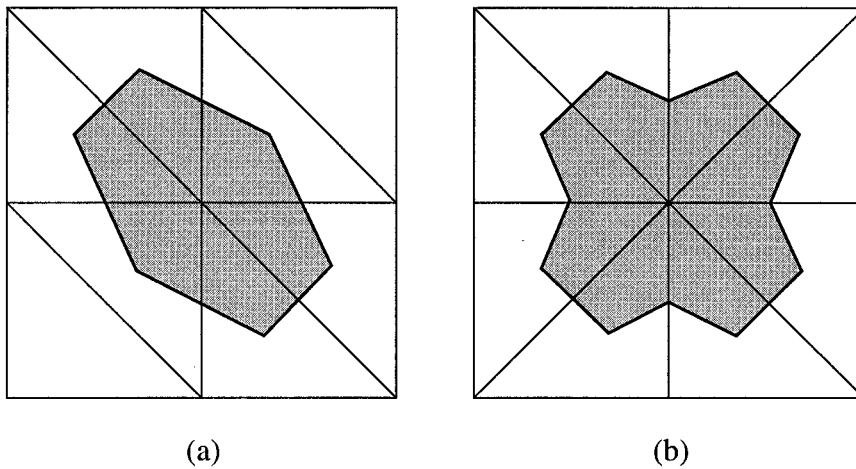


Figure 2. 2D nodal volumes for different mesh topologies

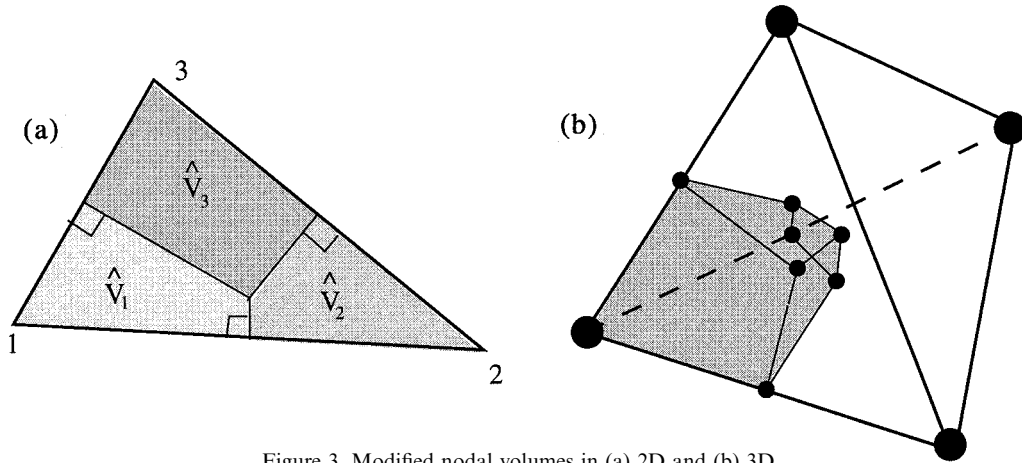


Figure 3. Modified nodal volumes in (a) 2D and (b) 3D

regular topology using only tetrahedra, implies that the nodal masses of any mesh can never be uniform, even when the nodes are equally spaced.

The lack of mass uniformity can be overcome by redefining the volume attached to a given node as the region in space formed by those points which are closer to the node in question than to any other mesh node. In the case of a triangle, the areas attached to each node can be obtained by joining the circumcentre to the mid-side points as shown in Figure 3(a). A similar but more complex geometrical procedure can be used for 3D cases. First the circumcentres of the four faces and of the tetrahedra itself (i.e. the centre of the sphere containing the four nodes) are evaluated. The volume nearest to a given node is then that of the hexahedra defined by the node itself, the circumcentre of the tetrahedron, the circumcentres of the three faces containing that node and the mid-side points of the three sides containing the node. This process is illustrated in Figure 3(b).

The resulting nodal masses and volumes are then evaluated by adding elemental contributions as

$$\hat{M}_a = \sum_{e=1}^{m_a} \rho_0^{(e)} \hat{V}_a^{(e)}, \quad \hat{V}_a = \sum_{e=1}^{m_a} \hat{V}_a^{(e)} \quad (5a,b)$$

Figure 4 shows that the nodal volumes evaluated in this way are only dependent on the nodal positions in space and are independent of the mesh connectivities. It is reasonable to assume therefore that this procedure will lead to a more realistic evaluation of the lumped nodal masses for a 3D tetrahedral mesh.

2.2. Large strain kinematics

Consider now the motion of a general body discretized using linear tetrahedral elements (see Figure 5). The motion is described by a mapping ϕ between the initial co-ordinates of each particle $\mathbf{X}$ and its current position $\mathbf{x}$ as

$$\mathbf{x} = \phi(\mathbf{X}) \quad (6)$$

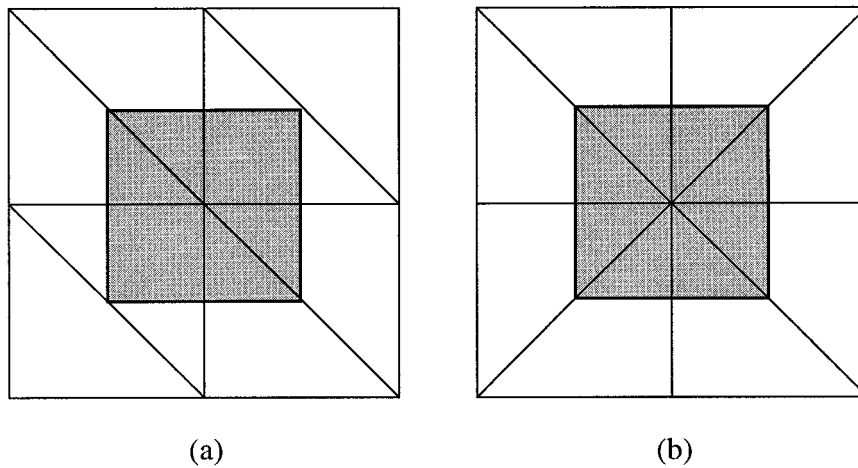


Figure 4. Modified nodal volumes for different mesh topologies

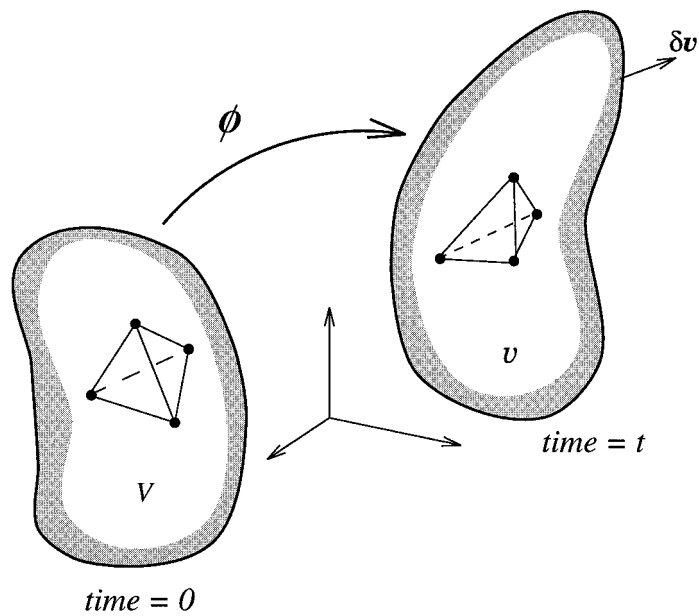


Figure 5. General body motion

Inside a given element e , the current geometry is interpolated from current nodal positions in the standard way as

$$\mathbf{x}^{(e)} = \sum_{a=1}^4 N_a \mathbf{x}_a \quad (7)$$

The derivative of the mapping ϕ defines the deformation gradient $\mathbf{F}$ which, with the help of equation (7), is evaluated as

$$\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}}, \quad \mathbf{F}^{(e)} = \sum_{a=1}^4 \nabla_0 N_a \otimes \mathbf{x}_a \quad (8a,b)$$

where $\nabla_0 N_a$ denotes the Cartesian derivatives of the shape functions at the initial configuration. The element formulation proposed below will rely on the ability to separate the volumetric component of the deformation gradient from its isochoric or distortional counterpart. For this purpose note first that the volume ratio is given by the determinant of $\mathbf{F}$ as

$$J = dv/dV = \det \mathbf{F} \quad (9)$$

and consequently the isochoric component of the deformation gradient, $\hat{\mathbf{F}}$, is¹⁰

$$\hat{\mathbf{F}} = J^{-1/3} \mathbf{F} \quad (10)$$

2.3. Strain energy function

Although the element formulation proposed is not restricted to hyperelastic materials, assume for simplicity the existence of a total elastic energy functional $\Pi(\mathbf{x})$ given as¹¹

$$\Pi(\mathbf{x}) = \int_V \Psi(\mathbf{F}) dV = \int_V \hat{\Psi}(\mathbf{F}) dV + \int_V U(J) dV \quad (11)$$

where Ψ is the strain energy density function with isochoric component $\hat{\Psi}$ and volumetric component U . In general, the isochoric strain energy must be a function of $\mathbf{F}$ through $\hat{\mathbf{F}}$. The simplest case is the so-called neo-Hookean material for which $\hat{\Psi}$ is¹⁰

$$\hat{\Psi}(\mathbf{F}) = \mu(\hat{\mathbf{F}} : \hat{\mathbf{F}} - 3) \quad (12)$$

The volumetric component U can only be a function of the volume ratio J , and the simplest and most commonly used case is¹¹

$$U(J) = \frac{1}{2} \kappa (J - 1)^2 \quad (13)$$

For compressible materials the coefficient κ represents a real material property known as the bulk modulus. In addition, in a penalty type of formulation, the same equation can be used to model incompressible materials by taking very large values of κ , typically $\kappa/\mu > 10^3$.

2.4. Standard volumetric internal forces

As a prelude to the modified formulation presented below, it is useful to derive the internal volumetric forces that result from the above volumetric strain energy for a standard linear tetrahedron. For this purpose, note first that, as both $\mathbf{F}$ and J are uniform over each element, the above integrals are easily obtained by adding individual elemental contributions as

$$\Pi(\mathbf{x}) = \sum_{e=1}^m \Pi^{(e)}(\mathbf{x}), \quad \Pi^{(e)} = V^{(e)} \hat{\Psi}(\mathbf{F}^{(e)}) + U(J^{(e)}) V^{(e)} \quad (14a,b)$$

where the value of J over an element e simply gives the ratio between the current and initial element volumes as

$$J^{(e)} = \frac{v^{(e)}}{V^{(e)}} \quad (15)$$

The directional derivative of the volumetric component of the strain energy functional in the direction of virtual velocity $\delta \mathbf{v}$ gives the volumetric virtual work component as¹¹

$$\delta W_{\text{vol}}^{(e)} = \mathbf{D}\Pi_{\text{vol}}^{(e)}[\delta \mathbf{v}] = \left. \frac{dU}{dJ} \right|_{J=v^{(e)}/V^{(e)}} V^{(e)} \mathbf{D}J^{(e)}[\delta \mathbf{v}] \quad (16)$$

Defining the element pressure $p^{(e)}$ as

$$p^{(e)} = \left. \frac{dU}{dJ} \right|_{J=v^{(e)}/V^{(e)}} = \kappa \left(\frac{v^{(e)} - V^{(e)}}{V^{(e)}} \right) \quad (17)$$

and differentiating $J^{(e)}$ in the direction of $\delta \mathbf{v}$ to give

$$\mathbf{D}J^{(e)}[\delta \mathbf{v}] = J^{(e)} \text{div } \delta \mathbf{v}^{(e)} \quad (18)$$

enables the volumetric virtual work component to be expressed as

$$\delta W_{\text{vol}} = \sum_{e=1}^m p^{(e)} v^{(e)} \text{div } \delta \mathbf{v}^{(e)} \quad (19)$$

The equivalent internal forces can now be derived by first expressing the virtual velocity inside the element in terms of nodal virtual velocities and shape functions, thus allowing the divergence term in the equation above to be expressed as

$$\text{div } \delta \mathbf{v}^{(e)} = \text{div} \left(\sum_{a=1}^4 N_a \delta \mathbf{v}_a \right) = \sum_{a=1}^4 \nabla N_a \cdot \delta \mathbf{v}_a \quad (20)$$

Substituting this expression into equation (19) and swapping the summation signs, on the basis that the nodal virtual velocities of a given node are common for all elements converging at that node, yields the volumetric component of the internal equivalent forces at node a as

$$\begin{aligned} \delta W_{\text{vol}} &= \sum_{e=1}^m \sum_{a=1}^4 p^{(e)} v^{(e)} \nabla N_a^{(e)} \cdot \delta \mathbf{v}_a \\ &= \sum_{a=1}^n \left(\sum_{e=1}^{m_a} p^{(e)} v^{(e)} \nabla N_a^{(e)} \right) \cdot \delta \mathbf{v}_a \\ &= \sum_{a=1}^n \mathbf{T}_{\text{vol},a} \cdot \delta \mathbf{v}_a \end{aligned} \quad (21)$$

where $\mathbf{T}_{\text{vol},a}$ is

$$\mathbf{T}_{\text{vol},a} = \sum_{e=1}^{m_a} \mathbf{T}_{\text{vol},a}^{(e)}, \quad \mathbf{T}_{\text{vol},a}^{(e)} = p^{(e)} v^{(e)} \nabla N_a^{(e)} \quad (22a,b)$$

It is well known that the above formulation leads to volumetric locking for even moderately large values of the Poisson ratio ν . This is due to the fact that, in effect, the volume of each element is required to remain constant, thereby enforcing m constraints in the motion of a mesh with at most $3n$ degrees of freedom. Inevitably, this number of constraints, combined with the fact that the ratio between number of elements and nodes is typically more than 5 to 1 for tetrahedral meshes, leads to volumetric locking.

2.5. Averaged nodal pressure formulation

To avoid the large number of constraints encountered in the standard formulation, the volumetric strain energy is approximated by assuming that the volume ratio J remains constant over the volume attached to each node, to give

$$\Pi_{\text{vol}}(\mathbf{x}) = \sum_{a=1}^n U(J_a) V_a \quad (23)$$

where the average nodal volumetric ratio is defined in terms of the current and initial nodal volumes as

$$J_a = v_a / V_a, \quad v_a = \sum_{e=1}^{m_a} \frac{1}{4} v^{(e)} \quad (24a,b)$$

Note that the simple volume definition is used here instead of the more complex nearest volume evaluation described in Figure 3(b). The corresponding algebraic derivations for the modified volume definition are complex and have not yet been obtained. However, the results obtained so far with the simple volume evaluation seem to be acceptable. It remains to be seen whether the modified volume definition will improve the accuracy of the pressure equations.

Starting from the above definition of the volumetric strain energy, the corresponding volumetric internal forces are obtained by first differentiating in the direction of arbitrary virtual velocities to give the volumetric internal virtual work as

$$\delta W_{\text{vol}} = \mathbf{D}\Pi_{\text{vol}}[\delta \mathbf{v}] = \sum_{a=1}^n p_a V_a \mathbf{D}J_a[\delta \mathbf{v}] \quad (25)$$

where the average nodal pressure is given as

$$p_a = \left. \frac{dU}{dJ} \right|_{J=v_a/V_a} = \kappa \left(\frac{v_a - V_a}{V_a} \right) \quad (26)$$

The directional derivative of the average nodal volumetric strain is obtained using the definition of nodal volumes given in equation (24(b)) together with equation (18) as

$$\begin{aligned}
 DJ_a[\delta \mathbf{v}] &= \frac{1}{V_a} Dv_a[\delta \mathbf{v}] \\
 &= \frac{1}{V_a} \sum_{e=1}^{m_a} \frac{1}{4} Dv^{(e)}[\delta \mathbf{v}] \\
 &= \frac{1}{V_a} \sum_{e=1}^{m_a} \frac{1}{4} V^{(e)} DJ^{(e)}[\delta \mathbf{v}] \\
 &= \frac{1}{V_a} \sum_{e=1}^{m_a} \frac{1}{4} v^{(e)} \operatorname{div} \delta \mathbf{v}^{(e)}
 \end{aligned} \tag{27}$$

This expression enables equation (25) for the volumetric component of the internal virtual work to be rewritten as

$$\delta W_{\text{vol}} = \sum_{a=1}^n \sum_{e=1}^{m_a} \frac{1}{4} p_a v^{(e)} \operatorname{div} \delta \mathbf{v}^{(e)} = \sum_{e=1}^m \bar{p}^{(e)} v^{(e)} \operatorname{div} \delta \mathbf{v}^{(e)} \tag{28}$$

where the order of summations has been swapped, thus allowing the average element pressure to be defined as

$$\bar{p}^{(e)} = \frac{1}{4} \sum_{a=1}^4 p_a^{(e)} \tag{29}$$

Note that this would correspond with the pressure at the centroid of the element that would be obtained from a linear interpolation from the nodal values.

Comparing equations (28) and (19) it is clear that the introduction of the virtual velocity discretization into equation (28), following the procedure explained in equations (20) and (21), will now lead to similar internal equivalent forces in terms of the average element pressure as

$$\mathbf{T}_{\text{vol},a}^{(e)} = \bar{p}^{(e)} v^{(e)} \nabla N_a^{(e)} \tag{30}$$

The proposed element, which is embodied in equations (24(a)), (26), (29) and (30), can be implemented into any existing dynamic code with very minimal additional expense. For nearly incompressible materials it only requires the nodal volumes to remain constant and hence only enforces n constraints. Given that there are in general $3n$ degrees of freedom, locking will not take place for sufficiently large meshes.

The nodal volumes used above are those given by the element centroids, as explained in Section 2.1. The use of the more complex nodal volume definition introduced in the above Section, based on the nearest points to a given node, has not yet been tested but offers an interesting alternative to the formulation presented above.

2.6. Multiple material interfaces

A final problem to be addressed is that of nodes on an interface between two different material types. In such cases it is not clear what bulk modulus should be used in equation (26). To circumvent this problem, a nodal volume is defined for each material type (i) converging at node a as

$$v_a^{(i)} = \sum_{e=1}^{m_a^{(i)}} \frac{1}{4} v^{(e)} \quad (31)$$

where $m_a^{(i)}$ represents the number of elements of material type (i) sharing node a . A different nodal pressure is then evaluated for each material using the above nodal volumes at the initial and final configurations, to give

$$p_a^{(i)} = \kappa^{(i)} \left(\frac{v^{(i)} - V_a^{(i)}}{V_a^{(i)}} \right) \quad (32)$$

In this way when the pressures are averaged over an element, only those corresponding to the right element material are now used.

3. EXAMPLES

This Section presents two elasto–plastic dynamic impact examples which are employed as benchmarks to validate the averaged nodal pressure tetrahedra developed above. The analysis of large-strain, large-deformation elasto–plastic behaviour is an ideal method of authenticating the performance of the tetrahedra, as the plastic deformation is entirely incompressible. As a consequence of this, any problems resulting from volumetric locking will immediately become apparent. Two different examples are addressed, namely a copper bar impacting against a rigid wall and an aluminium bar impacting against an inclined plane. Both were initially modelled using 8-noded hexahedra, and these results were compared against those obtained using tetrahedral meshes with both a standard constant pressure and the new averaged nodal pressure element formulations.

3.1. Impact of a cylinder against a rigid wall

This well-known example relates to the impact of a small cylindrical copper bar against a rigid planar wall. The bar has an initial length of 0.0324 m and initial radius 0.0032 m. The initial velocity of the bar is 227 m/s and the termination time of the problem is 80 μ s. Material properties for copper are defined in Table I. For the hexahedral mesh, the natural planes of

Table I. Copper bar material properties

Elastic modulus E	117 GN/m ²
Poisson ratio ν	0.35
Yield stress σ_y	0.4 GN/m ²
Hardening modulus H	0.1 GN/m ²
Density ρ_0	8930 kg/m ³

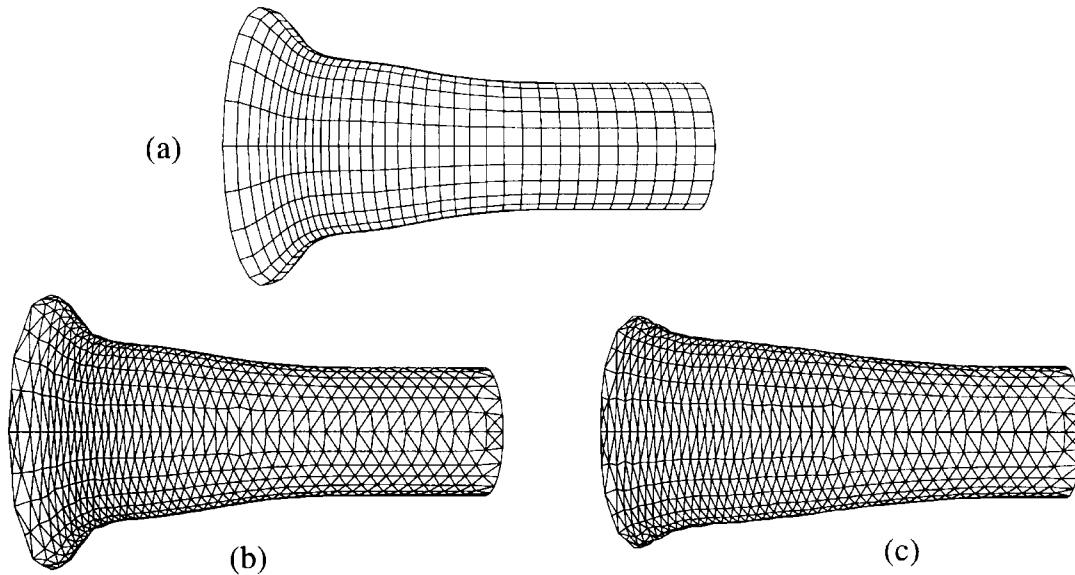


Figure 6. Final bar shape after 80 μ s using (a) hexahedral mesh, (b) new tetrahedra and (c) standard tetrahedra

symmetry were employed allowing only a quarter of the mesh containing 1369 nodes and 972 elements to be analysed. Approximately 6000 time steps were required to solve the problem, and the final shape is shown in Figure 6(a).

For the tetrahedral mesh, half the geometry was modelled due to the unstructured nature of the mesh generator employed. The half-mesh contained 1435 nodes and 5225 elements. The problem was initially analysed using standard tetrahedra and then with the new average nodal pressure formulation. Deformed shapes for both cases are shown in Figure 6(b),(c). Comparing these results with those obtained with the hexahedral mesh, volumetric locking is clearly observed for the standard element. In contrast, the deformed shape obtained with the new tetrahedra compare extremely well with those obtained from the hexahedral mesh. This is confirmed by comparison of the final radii of the impacting end of the bar, as shown in Table II.

3.2. *Oblique impact of an aluminium slender rod*

This Section presents the results from the numerical simulation of the impact of a cylindrical aluminium bar inclined 10° off the vertical axis against a horizontal rigid wall. The bar has an initial length of 0.305 m and initial diameter 0.00638 m. The initial velocity of the bar is 201.7 m/s and the termination time for the problem is 2.6 ms. The bar is modelled as

Table II. Copper bar — final radii

Element	Final radius, cm
Hexahedra	0.695
Standard tetrahedron	0.555
Proposed tetrahedron	0.699

Table III. Aluminium bar material properties

Bulk modulus κ	57.5 GN/m ²
Shear modulus G	26.5 GN/m ²
Yield stress σ_y	0.29 GN/m ²
Hardening modulus H	0.055 GN/m ²
Density ρ_0	2700 kg/m ³

elasto-plastic with isotropic strain hardening, and the material properties for aluminium are shown in Table III.

As with the previous example, the problem was initially modelled using 8-noded hexahedral elements, where by utilizing the natural plane of symmetry of the model only half the mesh was required for the analysis, containing 7503 nodes and 4656 elements. The tetrahedral mesh used contained a total of 7503 nodes and 23,280 elements. The problem was initially solved with the standard tetrahedra and then using the new element. The final shapes obtained in the three cases are shown in Figures 7(a)–(c). Clearly the standard tetrahedral mesh locks during the early stages

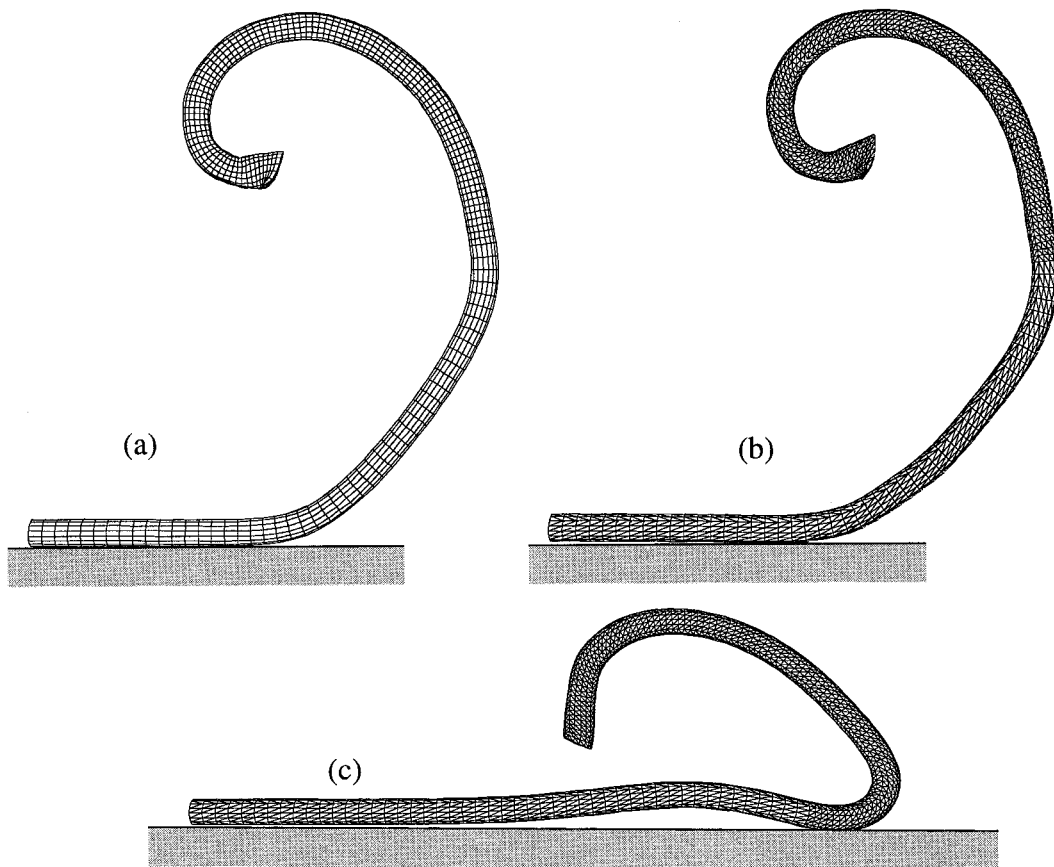


Figure 7. Final rod shape after 2.6 ms using (a) hexahedral mesh, (b) new tetrahedra and (c) standard tetrahedra

of the deformation and then follows a completely wrong solution path thereafter. However, the results obtained from the new tetrahedra compare extremely well with those obtained from the hexahedral meshes.

4. CONCLUSIONS

This paper has presented a simple average nodal pressure tetrahedron that can be implemented easily in explicit codes and is capable of simulating large nearly incompressible deformations without volumetric locking. Further research is still needed to evaluate the advantages of using different definitions of the nodal volumes. In addition it is important that the element is implemented in conjunction with a remeshing or mesh adaptation technique to exploit the full potential of the proposed methodology. However, it must be restated that the element is significantly more expensive than the standard hexahedron and hence should only be used when the initial geometry cannot be meshed with hexahedra or when a remeshing capability is imperative.

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